MANUFACTURING DATA JOURNEY

FROM DATA TO INSIGHTS

Transforming Industrial Data into Actionable Insights for Smarter Manufacturing Decisions

PRODUCTION | QUALITY | MAINTENANCE | INVENTORY | COST

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Industrial Data Analysis

SQL Server | Power BI | Python | Manufacturing Analytics

KEY PERFORMANCE INDICATORS
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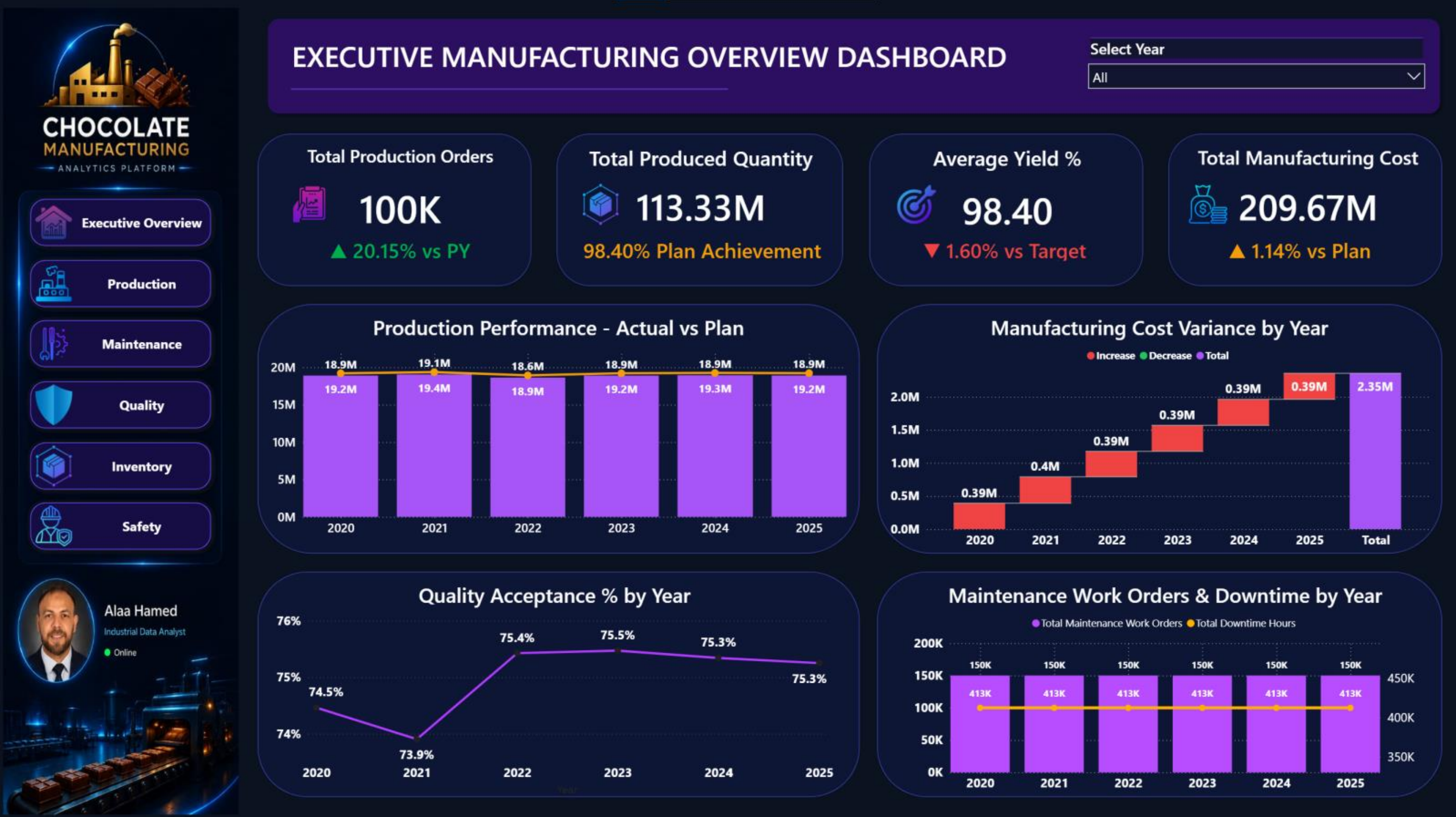
EXECUTIVE MANUFACTURING OVERVIEW

Management-Level Manufacturing Performance



KEY PERFORMANCE INDICATORS

1. Production Orders 100K

2. Produced Quantity 113.33M

3. Average Yield 98.40%

4. Manufacturing Cost 209.67M

EXECUTIVE INSIGHTS

What the Manufacturing Data Tells Us

01 PRODUCTION PERFORMANCE

98.40% Plan Achievement

Production reached 113.33M units, achieving 98.40% of the production plan.

Business Meaning:
Production performance is strong, but there is still a small gap to the planned output.

02 YIELD PERFORMANCE

1.60% Below Target

Overall yield reached 98.40%, remaining 1.60% below the target.

Business Meaning:
There is an opportunity to reduce production losses and improve process efficiency.

03 MANUFACTURING COST

1.14% Above Plan

Total manufacturing cost reached 209.67M, approximately 1.14% above plan.

Business Meaning:
Cost control should be monitored alongside production performance to protect manufacturing margins.

04 QUALITY PERFORMANCE

73.9% → 75.5%

Quality acceptance improved from approximately 73.9% in 2021 to around 75.5% in 2023, then remained relatively stable.

Business Meaning:
The improvement indicates better quality performance, while recent stabilization creates an opportunity for further optimization.

MANAGEMENT RECOMMENDATIONS

Turning Manufacturing Insights into Action

01 CLOSE THE PRODUCTION PLAN GAP

Production achieved 98.40% of the planned output.
Recommended Action:
Investigate the recurring gap between actual and planned production and identify the main operational constraints affecting output.
Focus Areas:
Production scheduling • Line capacity • Downtime • Material availability

02 IMPROVE YIELD PERFORMANCE

Overall yield reached 98.40%, remaining 1.60% below target.
Recommended Action:
Analyze production losses, scrap, rework, and process deviations to identify the main drivers of yield loss.
Focus Areas:
Scrap reduction • Process stability • Root Cause Analysis • Continuous Improvement

03 STRENGTHEN MANUFACTURING COST CONTROL

Manufacturing cost reached 209.67M, approximately 1.14% above plan.
Recommended Action:
Monitor cost variance by period, production line, product, and major cost drivers to identify opportunities for cost reduction.
Focus Areas:
Material cost • Production efficiency • Energy consumption • Maintenance cost

04 SUSTAIN QUALITY IMPROVEMENT

Quality acceptance improved from approximately 73.9% in 2021 to around 75.5% in 2023, then remained relatively stable.
Recommended Action:
Investigate the factors behind the improvement and establish continuous monitoring to prevent quality deterioration.
Focus Areas:
Quality defects • Process capability • Inspection results • Non-Conformance

MANAGEMENT PRIORITY

Close the performance gaps by connecting Production, Quality, Maintenance, and Cost data into one decision-making framework.

DATA → INSIGHT → ACTION → IMPROVEMENT

BUSINESS IMPACT & DECISION FRAMEWORK

From Data to Better Manufacturing Decisions

01 DATA VISIBILITY

Integrated manufacturing data provides management with a unified view of Production, Quality, Maintenance, and Cost performance.
Business Value:
One reliable view of operational performance instead of disconnected departmental reports.

02 PERFORMANCE MONITORING

Power BI dashboards transform operational data into interactive KPIs and performance trends.
Business Value:
Management can quickly identify performance gaps, monitor trends, and compare actual results against targets and plans.

03 ROOT CAUSE ANALYSIS

Combining production, quality, maintenance, and cost indicators creates a stronger foundation for investigating performance deviations.
Business Value:
Move from reporting what happened to understanding where and why performance changed.

04 DATA-DRIVEN DECISIONS

The analytical framework supports faster and more informed manufacturing decisions.
Business Value:
Turn industrial data into actionable insights that support productivity, quality improvement, cost control, and operational reliability.

DECISION FRAMEWORK

DATA
↓
KPI
↓
INSIGHT
↓
ACTION
↓
BUSINESS IMPACT

BUSINESS IMPACT AREAS

Productivity • Quality • Cost Control • Reliability • Continuous Improvement

TECHNOLOGY & PROJECT ARCHITECTURE

Building the Data Foundation Behind the Dashboard

01 DATA SOURCE

Industrial Manufacturing Data
Production • Quality • Maintenance • Inventory • Cost
Raw operational data generated across manufacturing departments.

02 DATA WAREHOUSE

SQL SERVER
Enterprise Data Warehouse designed to centralize and organize manufacturing data.
ETL Pipeline • Staging Layer • Master Layer • Data Validation

03 DATA MODEL

STAR SCHEMA
Fact tables connected with reusable Dimension tables to create a scalable analytical model.
Fact Production • Dim Date • Dim Product • Dim Equipment • Dim Plant

04 BUSINESS INTELLIGENCE

POWER BI

Interactive dashboards transform the analytical model into actionable manufacturing insights.
DAX Measures • KPIs • Interactive Filters • Drill-Through • Navigation

END-TO-END ANALYTICS FLOW

INDUSTRIAL DATA
↓
SQL SERVER DATA WAREHOUSE
↓
STAR SCHEMA DATA MODEL
↓
POWER BI SEMANTIC MODEL
↓
MANUFACTURING DASHBOARDS
↓
BUSINESS INSIGHTS
↓
DATA-DRIVEN DECISIONS

PROJECT OUTCOME

A scalable manufacturing analytics foundation that connects operational data with business decisions across Production, Quality, Maintenance, Inventory, and Cost.
TOOLS & TECHNOLOGIES

SQL Server • SQL • Power BI • DAX • Data Modeling • ETL • Manufacturing Analytics